UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS **5054/01**

Paper 1 Multiple Choice **May/June 2009**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

7338668226

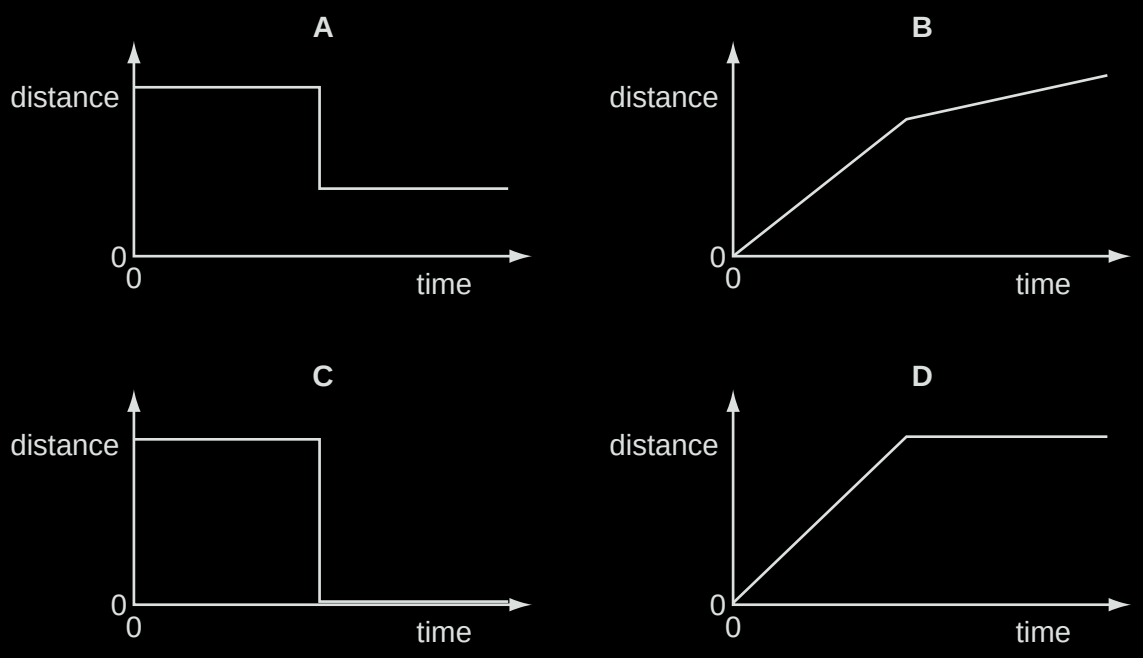
READ THESE INSTRUCTIONS FIRST

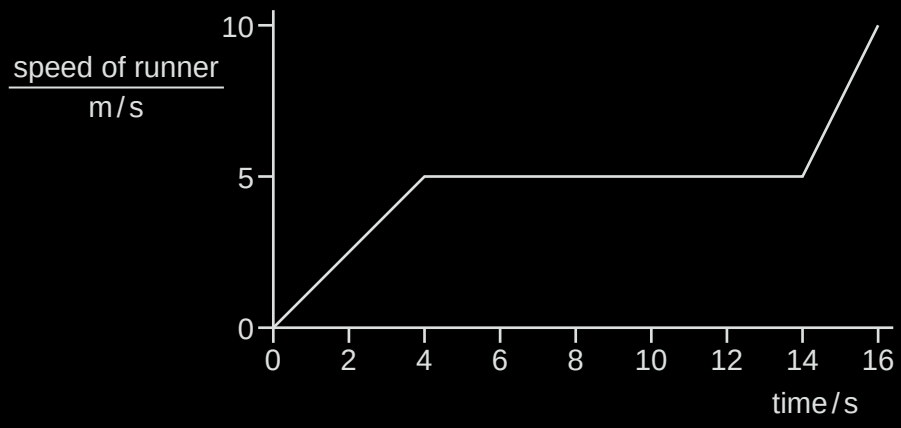
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

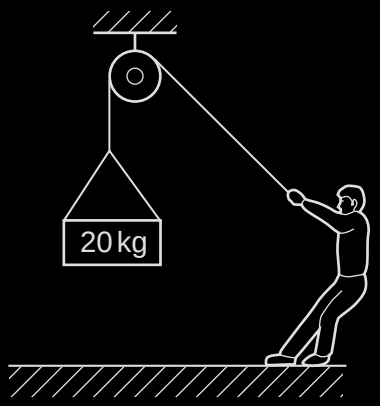

A
distance
0
0
time
B
distance
0
0
time
C
distance
0
0
time
D
distance
0
0
time



What is the distance travelled by the runner during the race?

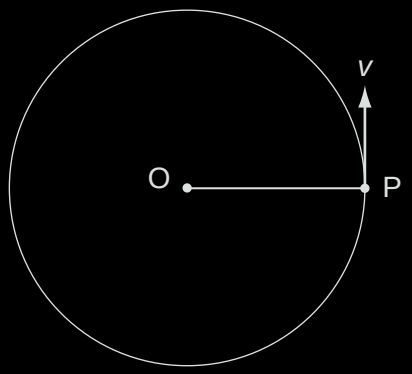
A 50 m **B** 65 m **C** 75 m **D** 90 m

5 A person just supports a mass of 20 kg suspended from a rope.



What is the resultant force acting on the mass?

A 0 N **B** 10 N **C** 20 N **D** 200 N



Which statement is true?

A A force of constant magnitude is acting in the same direction as *v*.

B A force of constant magnitude is acting towards O.

C The force on P varies in size as it moves around the circle.

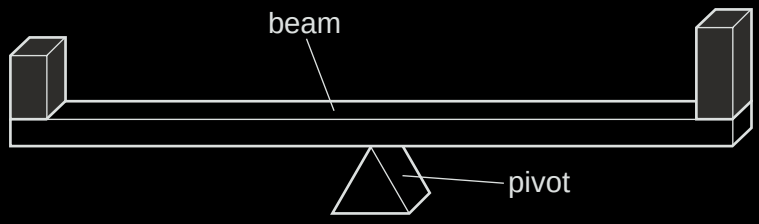
D There is no resultant force acting on P.

7 A measuring cylinder contains 118 cm^3 of water. When a small object is fully immersed in the water, the reading goes up to 132 cm^3. The object has a mass of 42 g.

What is the density of the object?

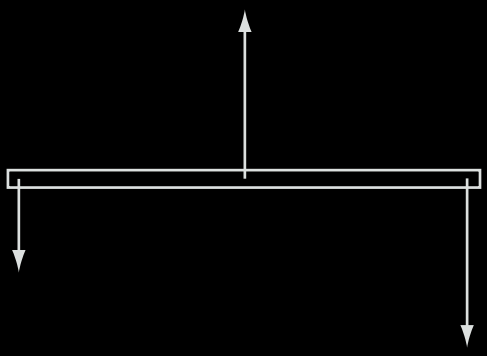
A $\frac{14}{42}\,g/cm^3$ **B** $\frac{42}{14}\,g/cm^3$ **C** $\frac{42}{118}\,g/cm^3$ **D** $\frac{132}{42}\,g/cm^3$

at its centre. The weight of the beam is negligible.

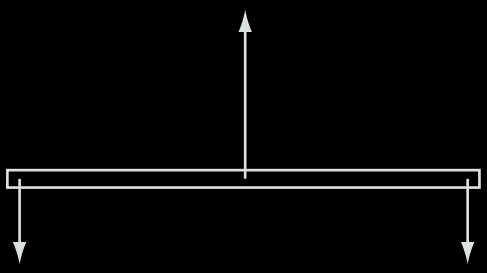


Which diagram shows the forces acting on the beam?
(The length of each arrow represents the size of a force.)

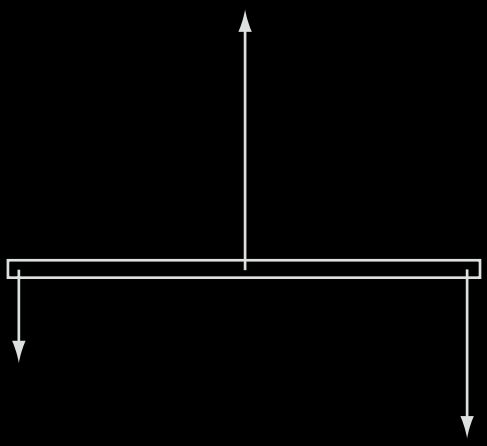
A



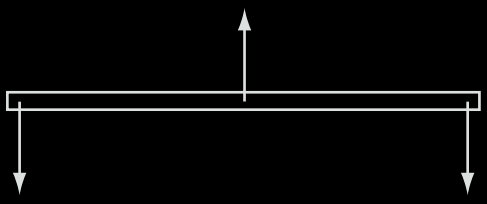
B

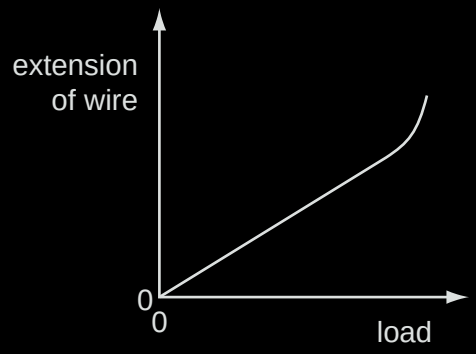


C



D





What does the graph show?

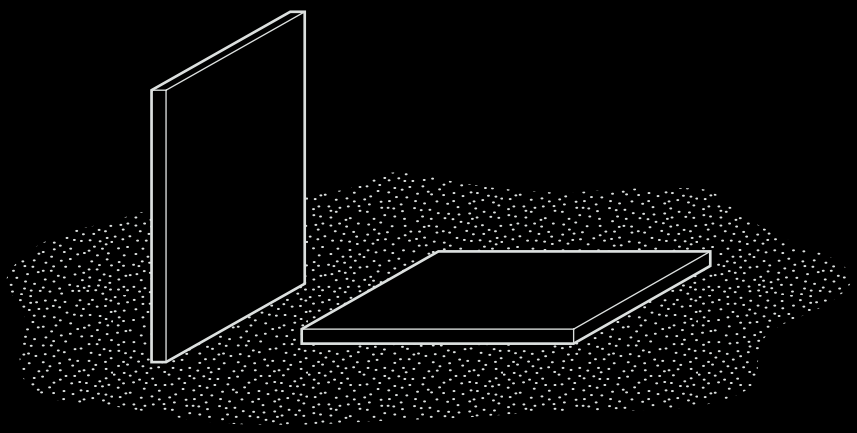
A At a certain load the wire becomes easier to extend.

B At a certain load the wire becomes harder to extend.

C The load and extension are directly proportional for any load.

D The load and extension are inversely proportional for any load.

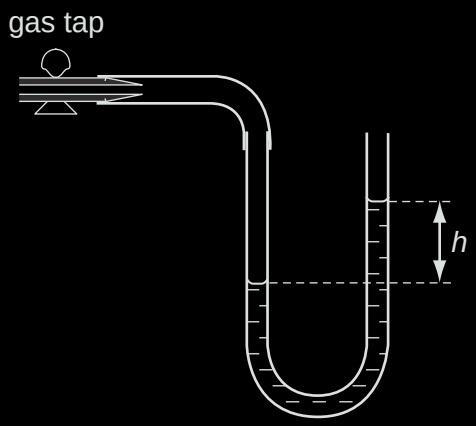
10 A builder leaves two identical, heavy, stone tiles resting on soft earth. One is vertical and the other is horizontal.



After a few hours, the vertical tile has started to sink into the soft earth, but the horizontal one has not.

Which row correctly compares the forces and the pressures that the tiles exert on the earth?

	forces	pressures
A	different	different
B	different	same
C	same	different
D	same	same



Which statement about the pressure of the gas supply is true?

A The pressure is *h* cm of water.

B The pressure is *h* cm of water below atmospheric pressure.

C The pressure is the same as atmospheric pressure.

D The pressure is *h* cm of water above atmospheric pressure.

12 Which process in the Sun produces energy?

A burning

B nuclear fission

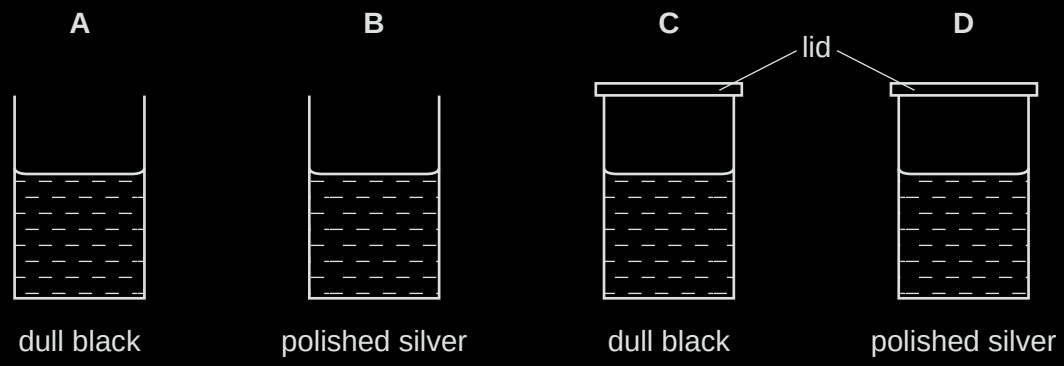
C nuclear fusion

D radiation

13 A crane lifts a load of 1000 N through a vertical height of 3.0 m in 10 s. The input power to the crane is 500 J / s.

What is the efficiency of the crane?

A 0.17 **B** 0.50 **C** 0.60 **D** 0.67



15 What makes a liquid-in-glass thermometer sensitive to a small change of temperature?

A a bulb with a thin glass wall

B a shiny liquid in its bore

C a stem with a thick glass wall

D a very narrow bore

16 A substance that is originally a solid is heated strongly for some time.

At one stage, the energy given to the substance is used as latent heat of vaporisation.

At this stage, what change does the energy cause?

A It breaks the bonds holding the molecules together. Molecules escape from the liquid.

B It breaks the bonds holding the molecules together. The solid becomes liquid.

C It makes the molecules move faster but there is still a strong attraction between them.

D It makes the molecules move faster and so the temperature rises.

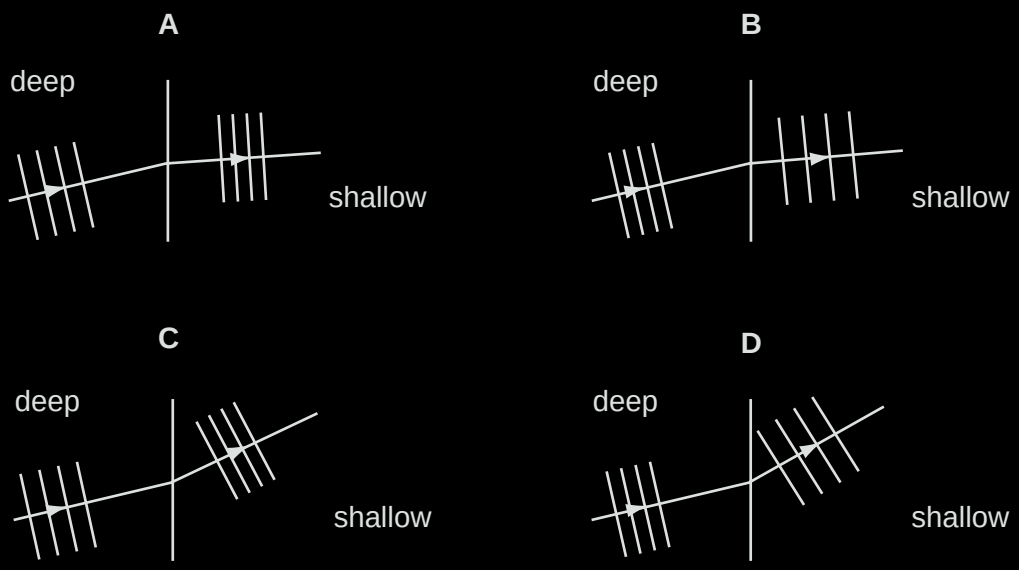
17 What happens when a metal bar is heated?

A The distance between the molecules increases, making the bar longer.

B The molecules get larger, making the bar longer.

C The molecules vibrate more quickly, making the bar denser.

D The speed of the molecules increases, making the bar thinner.



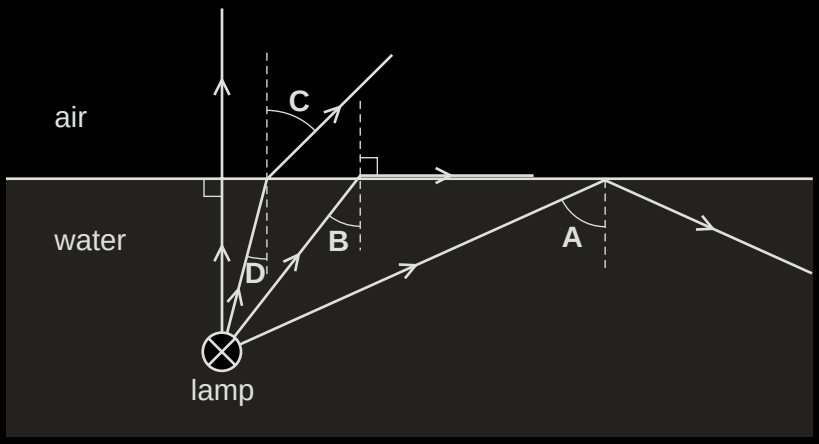
20 Which characteristics describe an image formed by a plane mirror?

A real and inverted

B virtual and upright

C real and larger than the object

D virtual and smaller than the object



22 When white light is dispersed by a prism, compared with blue light, the **red** light is

A slowed down less and refracted less.

B slowed down less and refracted more.

C slowed down more and refracted less.

D slowed down more and refracted more.

23 An ultrasonic tape-measure is used to find the distance to a wall. It sends out an ultrasonic pulse and times how long it takes for the reflected pulse to return from the wall.

The ultrasound has a frequency, a wavelength and a speed.

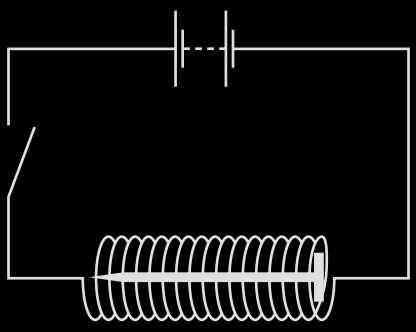
Which pair of values is needed to find the distance to the wall?

A frequency and wavelength

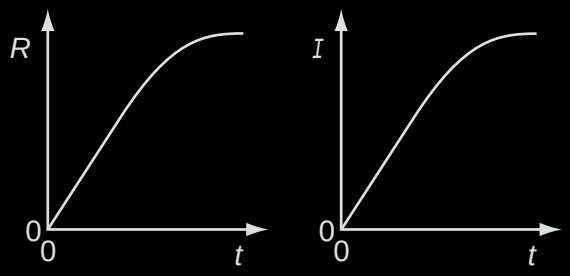
B frequency and time taken for the pulse to return

C speed and time taken for the pulse to return

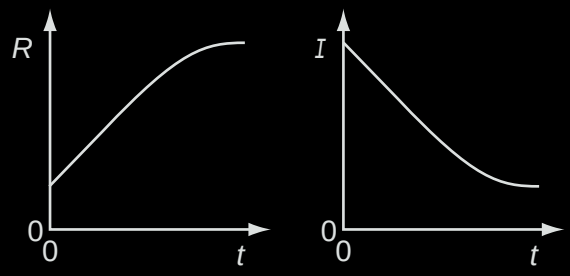
D wavelength and time taken for the pulse to return



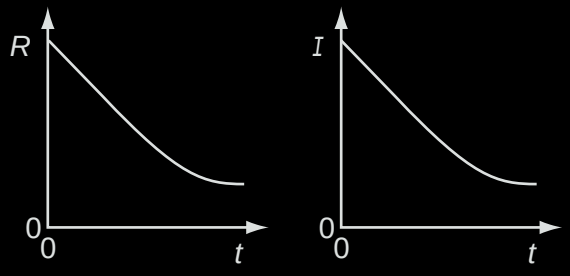
A



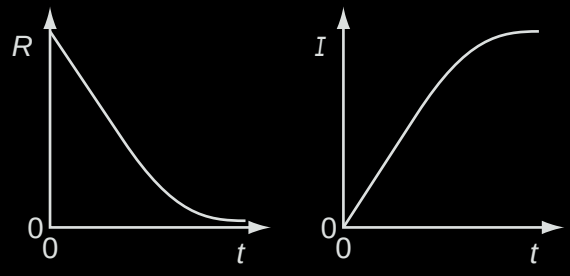
B



C



D



27 One volt can also be written as

A one coulomb per ampere.

B one coulomb per joule.

C one joule per ampere.

D one joule per coulomb.

28 A student has a chain of 20 lamps. These are wired in series and connected to the mains.

One lamp blows and all the others go out.

The student wants to find the faulty lamp and replace it.

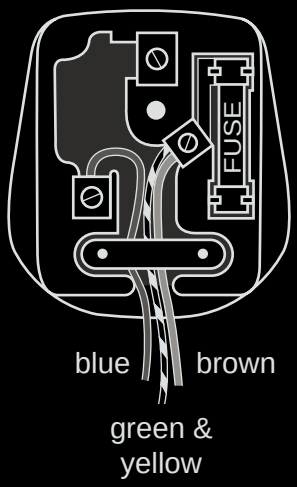
Where should the student begin?

A anywhere, because the current was the same in each lamp

B at the live end of the chain, because the current was greatest there

C at the middle of the chain, because the current was greatest there

D at the neutral end of the chain, because the current was least there



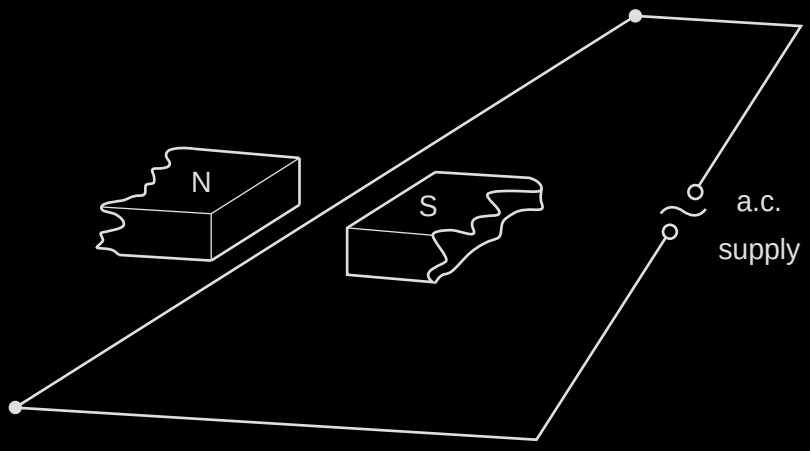
What is the effect of using the plug wired in this way?

A The fuse in the plug blows.

B The metal case is live.

C The neutral wire melts.

D The vacuum cleaner catches fire.



Which way will the wire move?

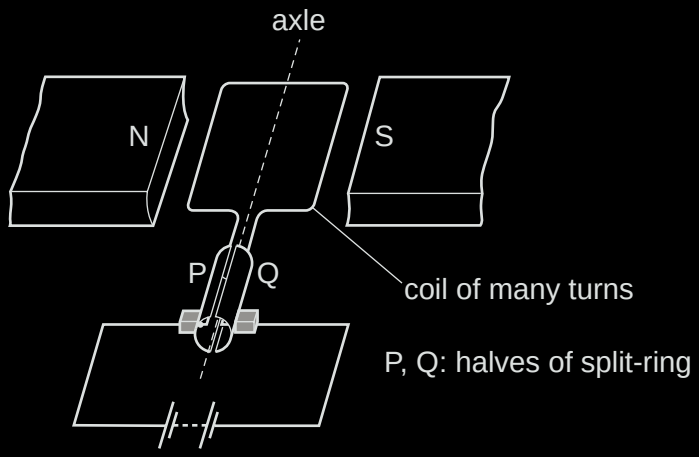
A left and right

B right only

C up and down

D up only

32 A d.c. motor consists of a coil of many turns rotating in a fixed magnetic field. The coil is connected to a d.c. supply through a split-ring commutator.

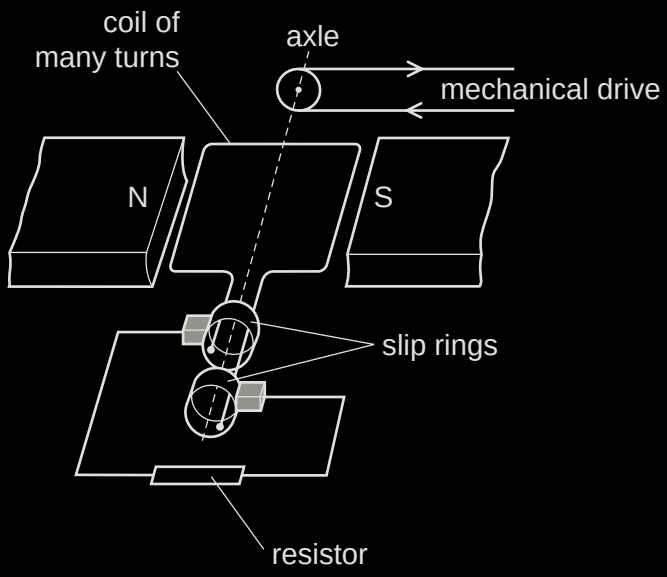


Some changes are made, one at a time.

- The d.c. supply is reversed.
- The coil is turned before switching on, so that P starts on the right and Q on the left.
- The poles of the magnet are reversed.
- The turns on the coil are increased in number.

How many of these changes make the coil rotate in the opposite direction?

A 1 **B** 2 **C** 3 **D** 4

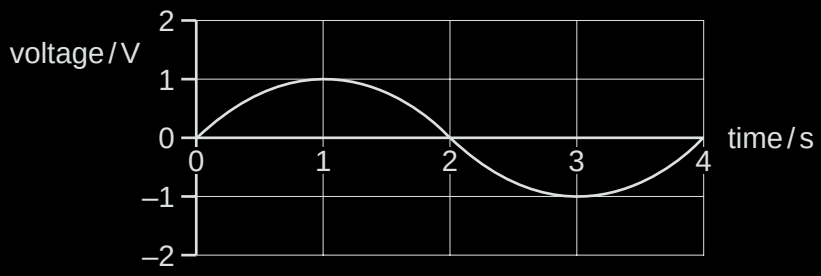


Some changes are made, one at a time.

- The speed of the drive is changed.
- The strength of the magnets is changed.
- The number of turns in the coil is changed.
- The value of the resistor is changed.

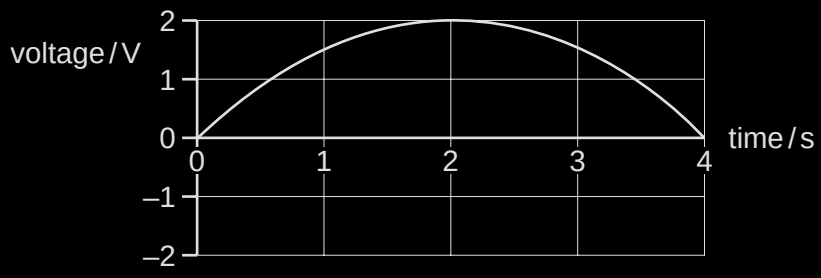
How many of these alter the value of the e.m.f. generated in the coil?

A 1 **B** 2 **C** 3 **D** 4

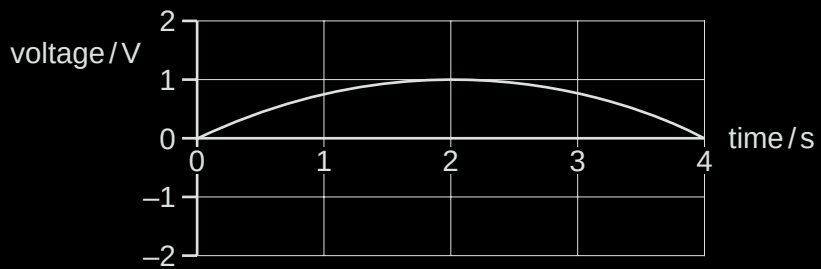


Which graph shows how the voltage varies with time when the generator rotates at twice the original speed?

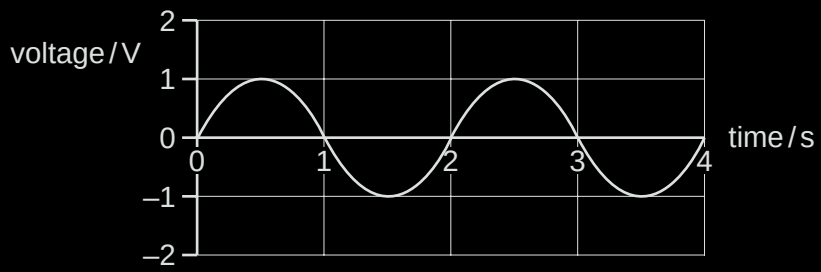
A



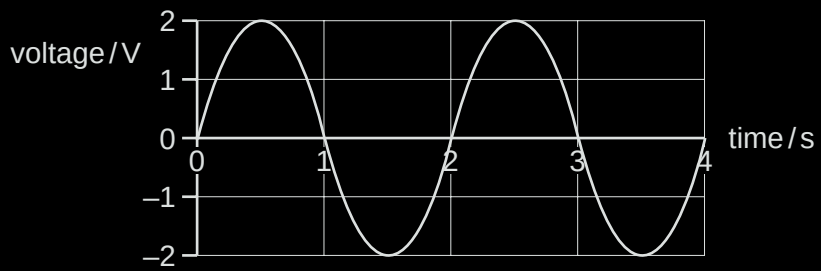
B

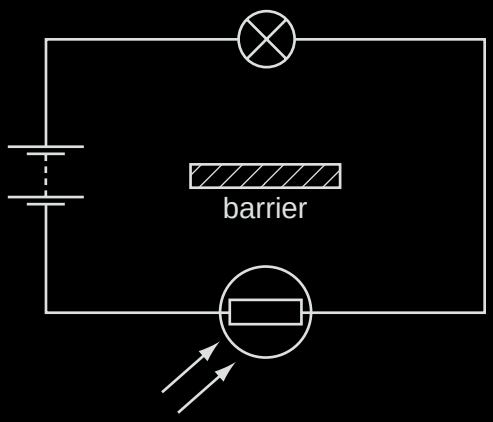


C



D





When low intensity light falls on the LDR, the lamp glows dimly.

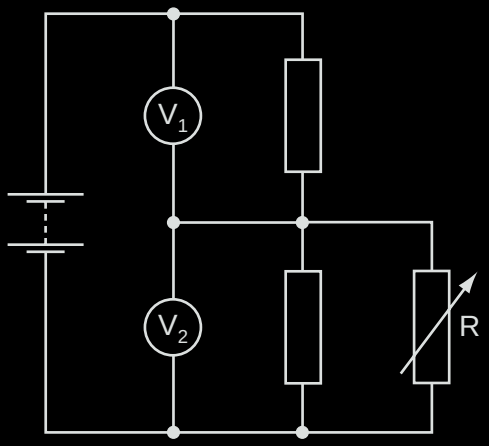
If the light intensity on the LDR increases, what happens to the lamp?

A It gets brighter.

B It gets dimmer but does not go out.

C It goes out.

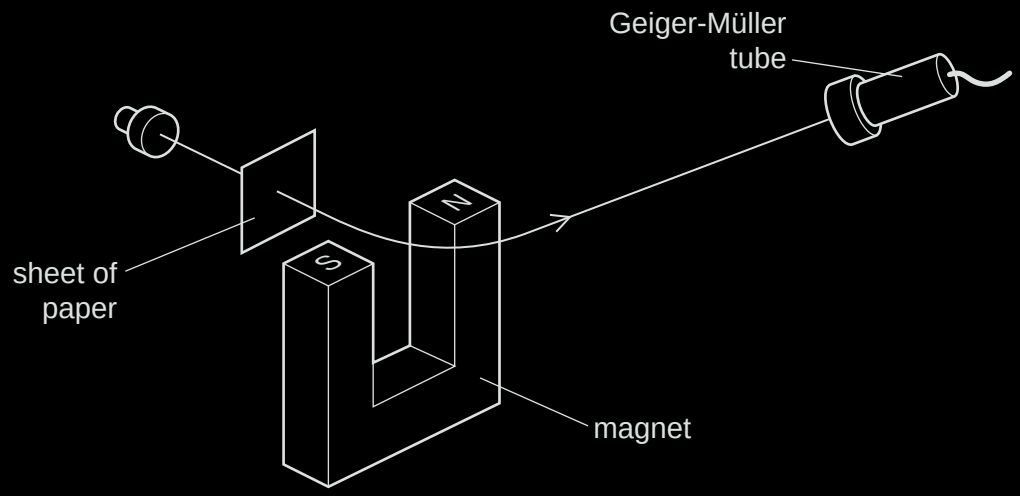
D It stays the same.



The resistance of R increases.

What happens to the two voltmeter readings?

	reading on V_1	reading on V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases



What is detected by the Geiger-Müller tube?

A alpha-particles and beta-particles

B alpha-particles only

C beta-particles and gamma-rays

D beta-particles only

39 The count-rate from a radioactive source falls from 400 to 50 in 3.0 minutes.

What is the half-life?

A 0.75 minutes

B 1.0 minutes

C 2.7 minutes

D 8.0 minutes

40 What are the numbers of neutrons, protons and electrons in a neutral atom of $^{235}_{92}U$?

	number of neutrons	number of protons	number of electrons
A	92	143	143
B	92	235	235
C	143	92	92
D	235	92	92